

Giovanni August Immanuel Wijaya

giovanniaugustw@gmail.com 089680029862 Giovanni August giovanniaugust

github.com/ghyoco portfolio

Profile

I'm a fifth-semester Computer Science undergraduate at Bina Nusantara University with practical experience in building computer vision pipelines and predictive machine learning models. Seeking an AI/ML Engineer internship to contribute to production ML work and grow as an engineer.

Education

SMAK Penabur Gading Serpong 07/2021 – 05/2024

Bina Nusantara University, *Computer Science* 09/2024 – present
GPA: 3.81

Projects

PathFinder, *Real-Time Lane Detection Using Computer Vision* 02/2026 – 06/2026

- Engineered a computer vision pipeline capable of detecting lane boundaries and calculating vehicle lane drift across lightings (day and night).
- Implemented Bird's-Eye View (BEV) inverse perspective mapping, sliding-window clustering, and 2nd-degree polynomial curve fitting ($x = ay^2 + by + c$).
- Designed a FastAPI backend and deployed a live interactive web app with Docker on Hugging Face Spaces.

Smart Building Time-Series Energy Forecasting, 02/2026 – 06/2026
Comparing LSTM vs XGBoost for energy load forecasting on smart buildings

- Built a forecasting pipeline on the CU-BEMS dataset to predict 30-minute building power loads from multi-floor energy and indoor air quality sensor data.
- Built lag, rolling window, and cyclical time features to benchmark LSTM networks against XGBoost using chronological TimeSeriesSplit cross-validation.
- Achieved an R^2 of 0.9818, SMAPE of 6.95%, and 68.1% RMSE reduction over the baseline with XGBoost, validated via Diebold-Mariano testing.

Volunteering Experience

Bimbingan belajar Sitanala, *Volunteer* 02/2026 – 04/2026
Taught basic math and English lesson to elementary students in Dutasia Sitanala Sitanala. Tangerang City

Certificates

Pelatihan Azure AI Fundamentals: Microsoft AI-900T00-A Belajar AI dari Dasar

Technical Skills

- Python
- Numpy
- pandas
- Docker
- OpenCV
- FastAPI
- scikit-learn
- TensorFlow
- Git / GitHub
- Matplotlib
- TypeScript
- MySQL
- Vercel

Languages

Indonesian — Native/Bilingual

English — Proficient